

The Bottom Boundary Layer of the Deep Ocean

LAURENCE ARMI AND ROBERT C. MILLARD, JR.

Woods Hole Oceanographic Institution, Woods Hole, Massachusetts 02543

Some aspects of the bottom boundary layer of the deep ocean are exhibited in profiles of salinity and temperature made with a Woods Hole Oceanographic Institution/Brown CTD microprofiler. Profiles from the center of the Hatteras Abyssal Plain have a signature that is characteristic of mixing up a uniformly stratified region. Over rough or sloping topography, to the east and west of the abyssal plain, the profiles have more complicated structure. All profiles show a well-mixed layer above the bottom; the penetration height of the layer varies from about 10 to 100 m and is correlated with the one-day mean velocity, inferred from current meters located above the bottom boundary layer. Over the Hatteras Abyssal Plain the mixed layer extends above the bottom about 6 times the turbulent Ekman layer height. Over rough and sloping topography the penetration height is between the Ekman layer height and the height observed on the flat abyssal plain.

1. INTRODUCTION

Profiles of salinity and potential temperature taken to the bottom of the deep ocean show a well-mixed region in both salinity and potential temperature just above the bottom; this bottom boundary layer is often referred to as the benthic boundary layer. An example of a salinity and potential temperature profile, in this case taken over the smooth Hatteras Abyssal Plain, is shown in Figure 1. (The traces from both lowering and raising of the profiler are shown.) Here the well-mixed region extended to about 55 m above the bottom and was bounded by a sharp interface across which the salinity, potential temperature, and potential density changed. Above the interface there was a nearly uniformly stratified region with constant potential temperature, salinity, and potential density gradients.

The profile shown in Figure 1 is typical of most profiles taken over the smooth Hatteras Abyssal Plain. However, profiles taken over regions with rough or sloping topography or at the edge of the abyssal plain have a more complicated signature. Although the signature is more complicated in these regions, the well-mixed layer at the bottom was always observed. This paper reports on preliminary experimental and theoretical investigations on the nature of the bottom boundary layer, our immediate interest being to understand the simple structure formed on the smooth Hatteras Abyssal Plain.

Profiles similar to those presented here are also presented by Amos *et al.* [1971] for the region of the Blake-Bahama Outer Ridge. Biscaye and Eittreim [1974] and Eittreim *et al.* [1975] show temperature, nephelometer, suspended particulate, and excess radon profiles over the Blake-Bahama Outer Ridge and the Hatteras Abyssal Plain; the profiles indicate that suspended material and excess radon are well mixed in the layer, which is also well mixed in potential temperature. Weatherly [1972] and Weatherly and Niler [1974] have made profiles of the bottom boundary layer beneath the Florida Current, and Millard [1974] has presented profiles taken between 28° and 40°N along 70°W and at the internal wave experiment (Iwex) site at 27°44'N, 69°51'W on the Hatteras Abyssal Plain.

We believe the bottom boundary layer plays an important role in the dynamics of mesoscale motions as well as in the absorption and reflection of internal waves. Previous dynamical studies have been made by Wimbush and Munk [1971],

Weatherly [1972, 1975], Thompson [1973], and Csanady [1974]. The importance of bottom boundary layer dynamics to the transport of sediment is evident in the collections of papers edited by Gordon [1972] and Gibbs [1974].

In the next two sections we describe various aspects of near-bottom profiles. In section 2 the profiles are categorized by the types of topographic features over which they were found. The thesis that the well-mixed region of the bottom boundary layer is a dynamic feature is supported in section 3 by correlations of the penetration height of the well-mixed layer with the velocity observed in the water column above the layer. The results of a survey of the spatial and temporal variability of the bottom boundary layer are reported in section 4. In section 5 we review theoretical models of the bottom boundary layer and find that none describe well the large penetration height observed.

2. DESCRIPTION AND CATEGORIZATION OF BOTTOM BOUNDARY LAYERS

The profiles of salinity and potential temperature were all made with a Woods Hole Oceanographic Institution (WHOI)/Brown CTD microprofiler. The technical details of the instrument are described by Brown [1974], and the method of calibration and data handling by Fofonoff *et al.* [1974]. The profiler was lowered to within 3 m (3 dbar) of the bottom as indicated to the winch operator by a bottom-sensing trip switch alarm described by Knox and Millard [1973]. All the profiles displayed in the figures here thus extend to within less than 3 m of the bottom. The CTD profiles were made primarily in conjunction with the setting and retrieval of current meters as part of the WHOI buoy group's participation in the Mid-Ocean Dynamics Experiment (Mode). A few profiles were made near current meter moorings during the experiment and are also included here. The profiles are here numbered or named by the mooring near which they were made; the locations of the moorings are shown in Figure 2a.

Both salinity and potential temperature were measured for each profile. It was found in all cases that within the absolute accuracy of measurement (salinity, $\pm 0.003\%$; temperature, $\pm 0.0015^\circ\text{C}$) the θ/S (potential temperature/salinity) relationship follows that found by Worthington and Metcalf [1961]. Because the resolution of potential temperature gradients (temperature, $\pm 0.0005^\circ\text{C}$; depth, ± 0.1 dbar) is approximately 1 order of magnitude better than salinity gradients and the good θ/S relationship exists, we have only shown the

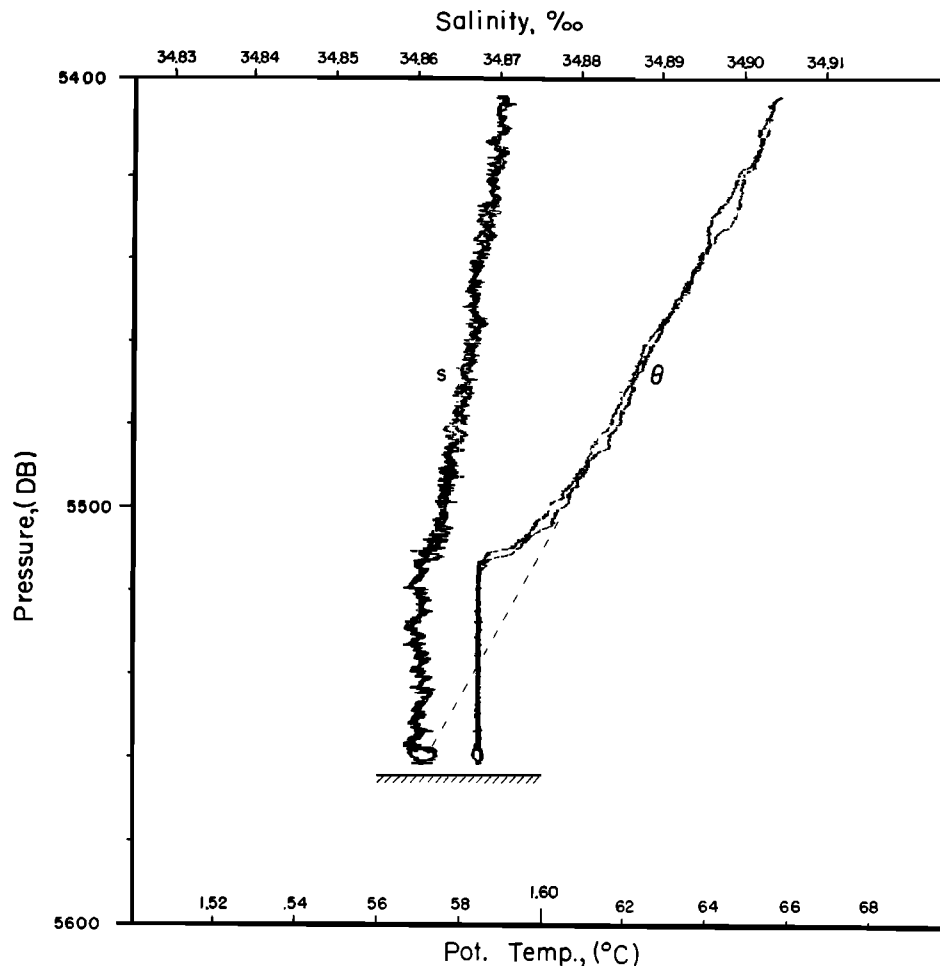


Fig. 1. A salinity, potential temperature profile made at mooring 3 (refer to Figure 2a) in the middle of the Hatteras Abyssal Plain. Dashed line indicates structure could have formed by mixing up the stratified region above. The traces from both lowering and raising of the profiler are shown.

temperature profiles in all the subsequent figures. In order for any profile to be included here the mooring near which any profile was made must have had an operational current meter at or beneath a nominal depth of 3000 m. (The current meter data are used later in section 3.) This has limited the number of profiles presented; however, the sample is typical of this region.

The well-mixed region of the bottom boundary layer is a ubiquitous feature and cannot be attributed to a distinctive water mass or pool. This is illustrated in Figure 2b, in which all the profiles considered in this paper, except those profiles from the Mode center and Mode east moorings which are included in Figure 2c, are superposed. Figure 2b also shows that the profiles fall into two distinct categories or types. Those taken over the smooth Hatteras Abyssal Plain (the profiles which extend deeper than 5500 dbar) generally have the same signature, the clearest example of which was used in Figure 1. The profiles taken over the neighboring sloping and rough topography exhibit more complicated structure. The distinction between the various profile types is best seen in Figure 2c, in which all the profiles are categorized by the topographic features over which they were taken and are plotted relative to the bottom depth and potential temperature. We now consider the types in order of their complexity.

The simplest signature occurs over the middle of the Hat-

teras Abyssal Plain. As is illustrated in Figure 1, it is a signature which could result from simply mixing a uniformly stratified region. Most of the profiles taken over the center of the Hatteras Abyssal Plain exhibit this type of signature. There are exceptions (the profiles from mooring 6) which exhibit a double-step structure that may be due to erosion of a former sharp interface. Nonetheless, all the signatures could conceivably have been formed by mixing a former uniform gradient that extends to the bottom, as illustrated by the dashed line in Figure 1.

Profiles taken near the edge of the Hatteras Abyssal Plain at the Mode center mooring exhibit some of the simple structure found in the center of the abyssal plain; however, some of the profiles exhibit the complicated structure found further east at the Mode east mooring and other eastern moorings, 9, 15, and 16. The progression from the simple signature over the center of the Hatteras Abyssal Plain to the edge of the abyssal plain and into the rough topography is illustrated in Figure 2c.

We have also included in the topographic type profiles two taken at mooring 13, west of the Hatteras Abyssal Plain over the Blake-Bahama Outer Rise beneath the steady Western Boundary Undercurrent. These profiles, unlike the other profiles, could not have been formed by simply mixing up the stratified region that lies above. The lower portion of the well-mixed layer appears to have been advected away.

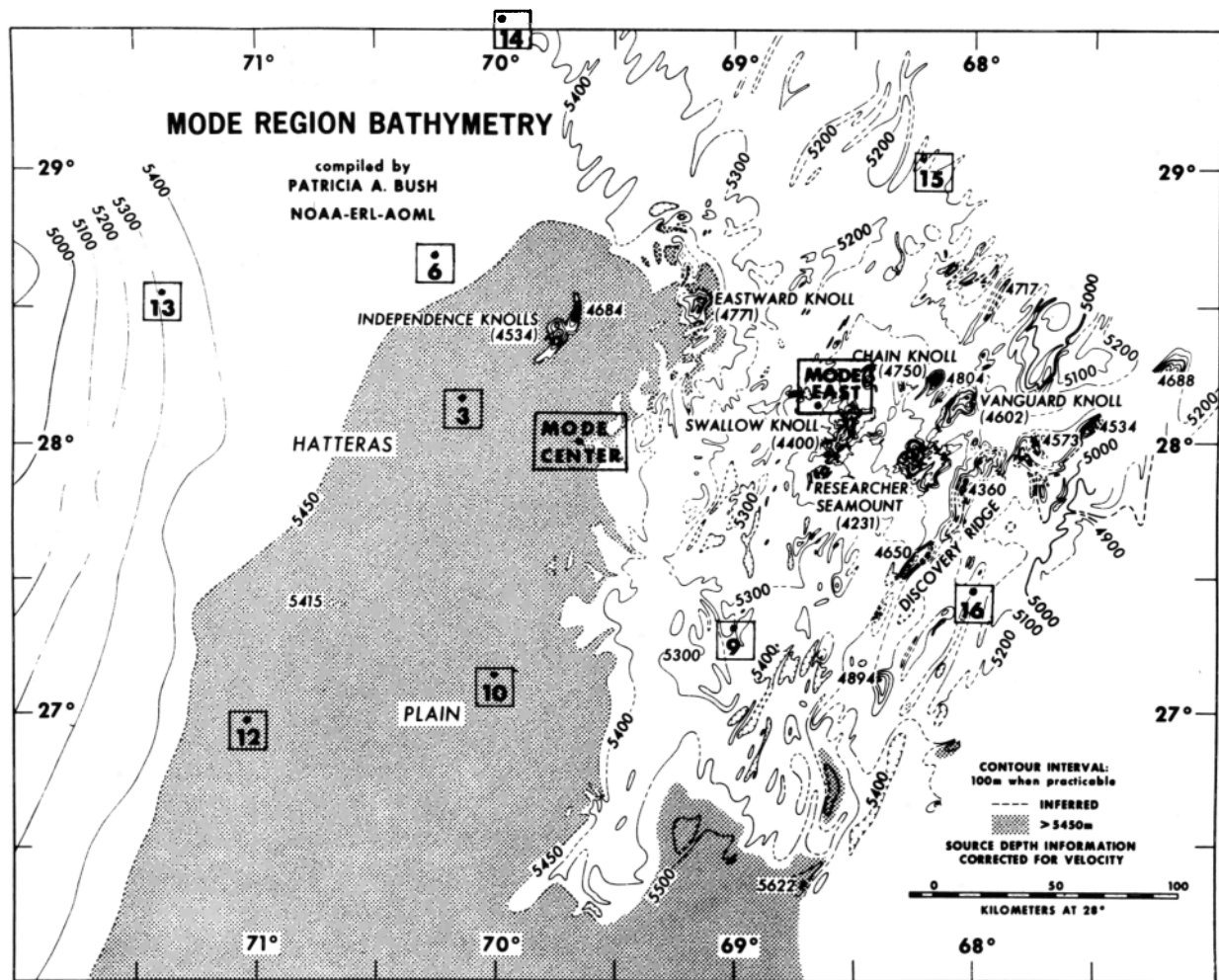


Fig. 2a. Locations of the current meter moorings near which the CTD profiles were taken. Bathymetry is from Bush [1975].

3. CORRELATION OF PENETRATION HEIGHTS WITH VELOCITY

The signature of the bottom boundary layer, with its well-mixed region bounded by an interface and a stratified region above, is distinctive. A signature like that of Figure 1 is also seen in laboratory experiments in which a uniformly stratified fluid in a tank is mixed up by stirring with a grid as in the experiments of Cromwell [1960] and Linden [1975] or a surface stress as in the experiments of Kato and Phillips [1969]. The signature is typical of a mixing process, the penetration of which is bounded by the interface which forms as a result of mixing the uniformly stratified fluid.

We believe the energy source for the mixing of the bottom boundary layer of the deep ocean is likely to be associated with the most energetic motions in the deep ocean. In the region of the smooth Hatteras Abyssal Plain, where the Coriolis parameter and the Brunt-Väisälä frequency in the stratified region above the mixed layer are nearly constant, a correlation should then exist between the penetration height of the well-mixed region and a suitably defined velocity. The result of such a correlation of penetration height and velocity is presented in Figure 3a for CTD profiles taken over the smooth topography of the Hatteras Abyssal Plain.

The penetration heights of the well-mixed layers are measured from the bottom to the beginning of the interface, as

estimated by eye. The velocity used for this correlation was the one-day mean velocity; all fluctuations, primarily tidal and inertial, with a period of a day or less are averaged. The one-day mean velocity is due to the most energetic motions in the deep ocean, i.e., the mesoscale motions. It is also the only velocity which will be nearly constant between 3000 m and the bottom; we are forced, because of a limited data base, to infer velocities near the bottom from current meters situated in some cases considerable distances above the bottom. The importance of velocities at inertial and higher frequencies is not fully understood; an estimate of their effect is given in section 5.

Records used are from those current meters located closest to the bottom and less than 7 km from the location of the CTD profile. The current meters were nominally at depths of 3000, 4000, and 100 m off the bottom (approximately 5400 m); the depths are also indicated on the correlation. When available, the velocity record used for the mean velocity was over the day preceding the time of the CTD profile. Since the CTD profiles were mostly taken in conjunction with setting and retrieving of the current meter moorings, the mean velocity is that for the one-day period after the time the CTD profile was made, if the profile was made in conjunction with setting of any mooring.

The validity of using velocity information from current meters placed up to 7 km from the location of the CTD profile

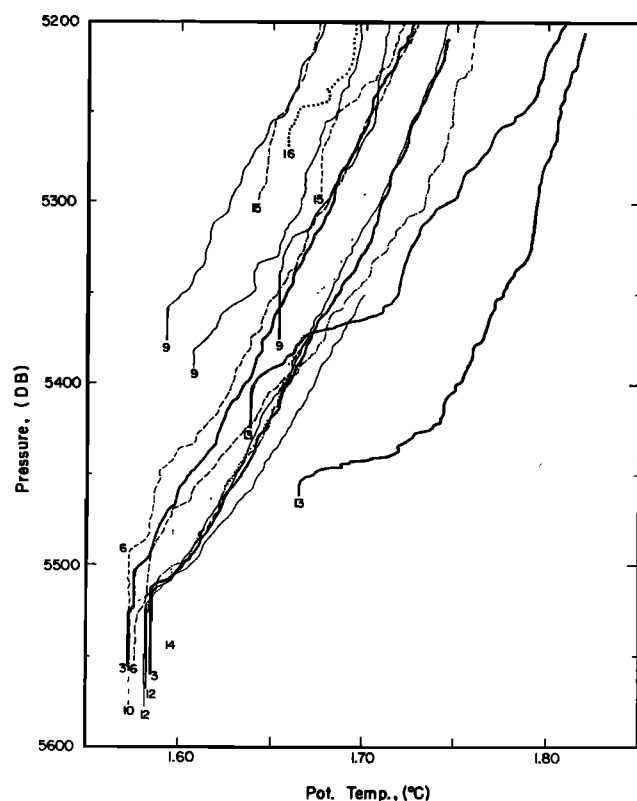


Fig. 2b. Potential temperature profiles, here numbered by the current meter mooring near which each was taken. Refer to Figure 2a for locations.

can be inferred from Gould *et al.* [1974]. At a depth of 4000 m a coherence of 0.95 was obtained for low-frequency fluctuations, of the type considered here, for a separation of 7 km. We refer to the velocity profiles of Sanford [1975] and the current meter records presented by the *Mode I Dynamics Group* [1975, p. 223, Figure 4.1a] for the validity of using current meters lo-

cated at depths of 4000 and 3000 m to infer mean velocities near the bottom.

The correlation of Figure 3a suggests that the well-mixed region of the bottom boundary layer is a dynamic feature; the penetration height is approximately proportional to the one-day mean velocity. It will be shown in section 5 that because of the considerable height of the mixed region, approximately 6 times the estimated Ekman layer height, the feature is probably advected along with the mean flow above. We now mention that the double-step structures found at mooring 6 and in some of the profiles at the Mode center mooring are found with a decreasing mean velocity. These structures are probably the signatures of eroded thicker layers associated with an earlier higher mean velocity.

A correlation (Figure 3b), similar to Figure 3a, has been prepared for CTD profiles taken over sloping topography in the Mode region neighboring the Hatteras Abyssal Plain. For sloping areas the correlation indicated for smooth areas appears to still hold; however, penetration heights, for any velocity, are in general less than those for the smooth areas. The data of Amos *et al.* [1971, Table 3] taken over the Blake-Bahama Outer Ridge suggest a similar correlation between the penetration of the well-mixed region and velocity estimated geostrophically. The scatter is large, however, when their geostrophic estimate is used to infer velocities near the bottom. Our estimate of velocities near the bottom over sloping areas, using current meters located above the bottom at 3000 and 4000 m, will not be as good as that for smooth areas; the correlation we have presented in Figure 3b must be viewed with this in mind.

Millard [1974] has suggested that a correlation may exist between the thickness of the layer and its potential temperature: cold layers are thick, while warm layers are thin. This is in contrast to what one would expect due to mixing of a uniform potential temperature gradient proposed here. The correlation found by Millard may be due to variations of potential temperature at any depth being a function of the velocity field. The extent of these variations can be seen in

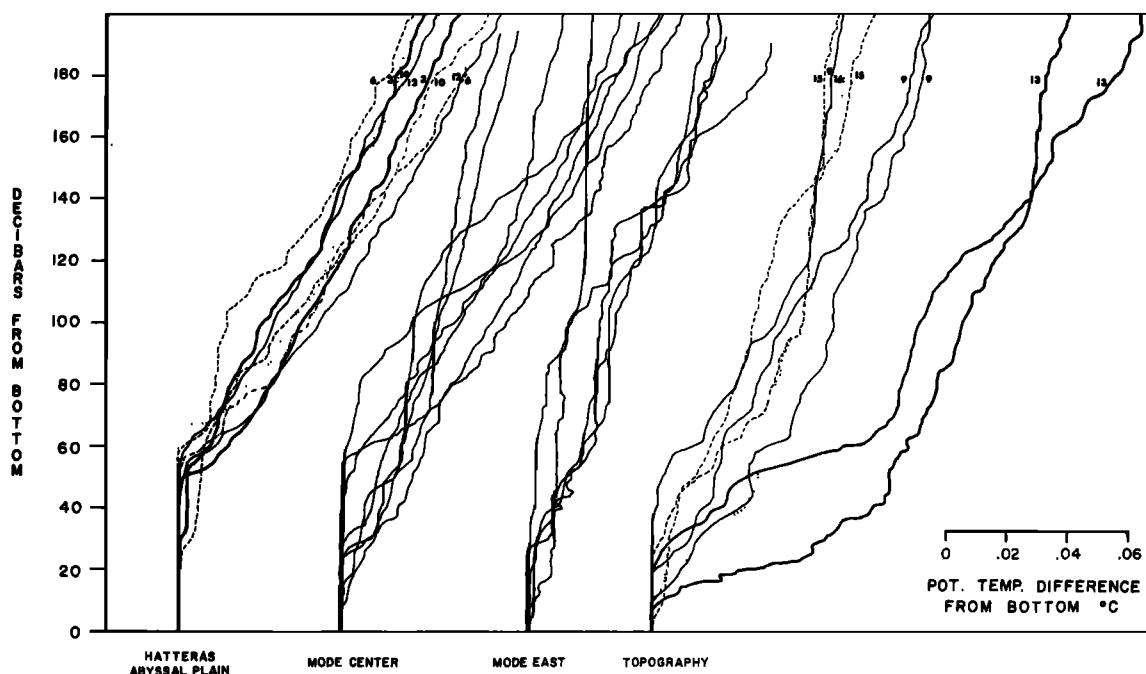


Fig. 2c. Potential temperature profiles plotted relative to the bottom in four groups. Refer to Figure 2a for locations.

Figure 2b and is up to 0.050°C . *Weatherly and Niller* [1974] also do not see as clear a correlation as *Millard* [1974] and point out instances for which this rule of thumb does not apply. The present data do not confirm the conjecture of *Millard* [1974].

4. SPACE AND TIME SCALES

A few profiles are discussed here, which give a preliminary picture of the spatial and temporal variability of the bottom boundary layer. The profiles were all made between July 26 and 29, 1974, in a region centered near the Mode center mooring at the eastern edge of the Hatteras Abyssal Plain. The locations of the profiles, which are here numbered as logged, are indicated in Figure 4a, which includes the topographic features of the area as compiled by *Bush* [1976]. Also shown are the locations of two current meter moorings, 538 and 542; the CTD stations were made in conjunction with the retrieval of mooring 538 and the setting of mooring 542. Each of the moorings had a deepest current meter, a Geodyne 850, at a nominal depth of 4000 m.

The potential temperature profiles are presented in chronological order in Figure 4b. All profiles made within the 1-km radius circle centered at $28^{\circ}03.6'\text{N}$, $68^{\circ}39'\text{W}$ are represented with solid lines; the remaining profiles are dashed. The profiles for this space-time survey exhibit both the simple signature found at the center of the abyssal plain as well as the more complicated signatures. This appears to be typical of this region at the edge of the abyssal plain. Variations in profiles are here found to occur over length scales as small as 10 km. Compare, for example, the double-step structure of profile 7 with the simple signatures of profiles 3, 4, and 6. Compare also the apparently eroded structure of profiles 16 and 17 with the simple signature of profiles 11, 12, 13, and 18. A profile similar to profiles 16 and 17 was also found at mooring 6 (see Figures 2b and 2c) and may be the signature of a decaying layer.

A short time series is also included here as profiles 9a–9e. Although the series was made as a time series, it also includes some spatial variability due to ship drift. This series shows a consistent penetration height of the well-mixed region but significant variations to the structure of the interface between the well-mixed layer and the stratified region above.

Profiles taken within a 1-km radius circle, drawn as solid lines in Figures 4b and 4c, show a growth of the well-mixed region from 20 to 68 m during the 3-day time period of the survey. We have exhibited this growth on Figure 4c superimposed on the current meter records from the two moorings. These records are unfiltered and contain fluctuations at inertial and higher frequencies, which at 4000 m have little correlation to similar fluctuations at the bottom. Unfortunately, no continuous record exists at one location; we have therefore pieced together the records of the nominal 4000-m current meters of two moorings separated by 10 km. The intent here is to indicate something of the temporal and spatial variability of the bottom mixed layer.

5. SOME QUANTITATIVE RELATIONSHIPS

Quantitative data characterizing the observed bottom mixed layers and the adjacent stratified fluid are as follows. The correlation of penetration height and velocity shown in Figure 3a for the smooth Hatteras Abyssal Plain gives

$$h/U \approx 1.2 \times 10^3 \text{ s} \quad (1)$$

typical values of h and U being 50 m and 4 cm/s. The celerity

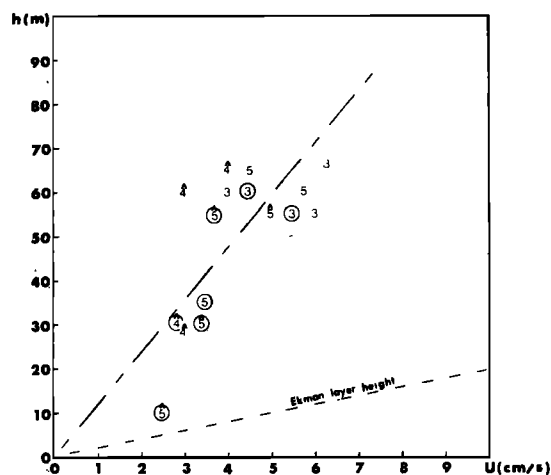


Fig. 3a. Correlation of penetration height of the well-mixed layer with velocity for stations on the Hatteras Abyssal Plain. Nominal depths of deepest current meters (3 = 3000 m, 4 = 4000 m, and 5 = 100 m off bottom) are indicated. A data point is circled if velocity average was taken over one-day period preceding time of profile. Stations made at the Mode center mooring are indicated by circumflex.

of long interfacial waves can be calculated by knowing the Brunt-Väisälä frequency in the stratified region above the mixed layer. The Brunt-Väisälä frequency,

$$N = (g \partial \rho / \rho \partial z)^{1/2} \quad (2)$$

is 0.4 cph ($7 \times 10^{-4} \text{ s}^{-1}$) and is nearly constant for all the profiles taken over the center of the Hatteras Abyssal Plain (see Figures 2b and 2c). By continuity of mass the reduced gravitational acceleration for the mixed layer is given by

$$g' = g \frac{\Delta \rho}{\rho} = \frac{1}{2} N^2 h \quad (3)$$

Neglecting the stratification above the interface, the celerity of a long wave on the interface is

$$c = (g' h)^{1/2} \quad (4)$$

about 2.5 cm/s for the typical penetration height of 50 m. An

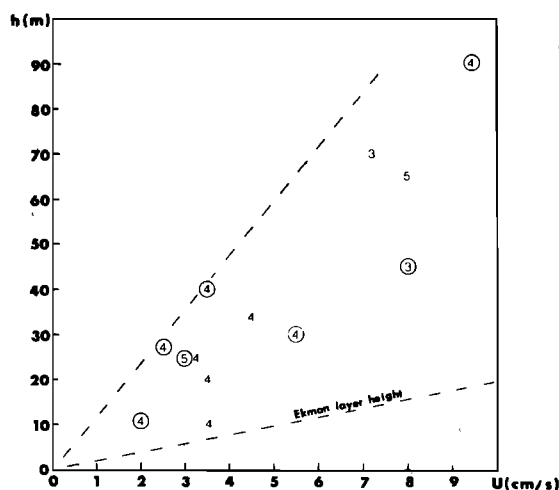


Fig. 3b. Correlation of penetration height of the well-mixed layer with velocity for stations taken over rough and sloping topography. Refer to Figure 3a for explanation of symbols.

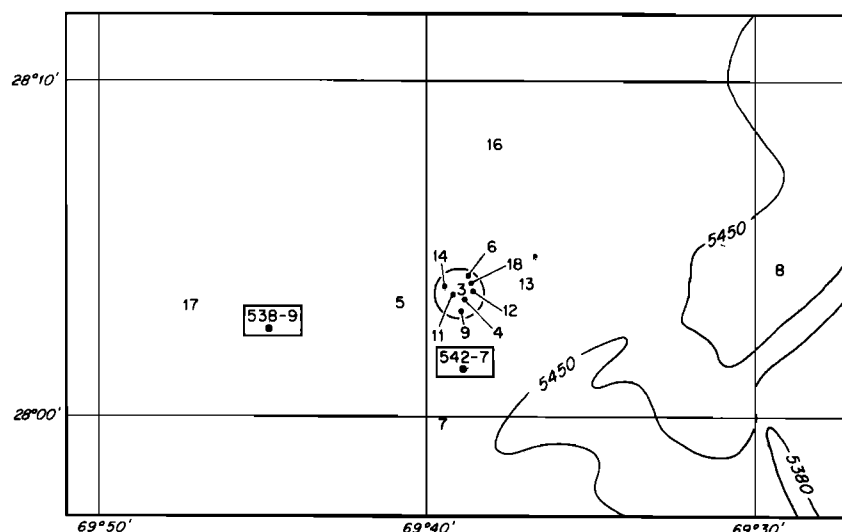


Fig. 4a. Locations of the profiles displayed in Figures 4b and 4c. The profiles are here numbered as they were logged. Topographic features from *Bush* [1976].

internal Froude number F can be defined for the bottom mixed layer [cf. *Turner*, 1973, p. 12]:

$$F = U/c \quad (5)$$

Using (3) and the ratio of penetration height to velocity, we find $F \sim 1.7$.

Can the bottom mixed layer we observed be treated as a classical turbulent Ekman layer? Such a treatment has been suggested by *Wimbush and Munk* [1971] and *Weatherly* [1972, 1975]. The experiments of *Caldwell et al.* [1972] and *Howroyd and Slawson* [1975] give the height of the turbulent Ekman layer h_e as

$$h_e = 0.4 (u_*/f) \quad (6)$$

(For the experiments of *Caldwell et al.*, $h_e \equiv \delta_{99}$, the height at which the velocity is 99% of the geostrophic velocity; for the

experiments of *Howroyd and Slawson*, h_e is defined as the height at which the velocity is parallel to the geostrophic velocity.) *Biscaye and Eittreim* [1974] report that photographs on the Hatteras Abyssal Plain display a monotonous flat mud bottom showing only some 'lebensspuren' with relief of about 1 cm. With the Coriolis parameter $f = 7 \times 10^{-5} \text{ s}^{-1}$ and the friction velocity u_* for a smooth bottom given by [cf. *Csanady*, 1967]

$$u_* = (1/30)U \quad (7)$$

the Ekman layer height to velocity ratio is

$$h_e/U \approx 2 \times 10^2 \text{ s} \quad (8)$$

Using the empirical result of (1), obtained from Figure 3a, we see the penetration height of the mixed layer is about 6 times the turbulent Ekman layer height. The Ekman layer can thus

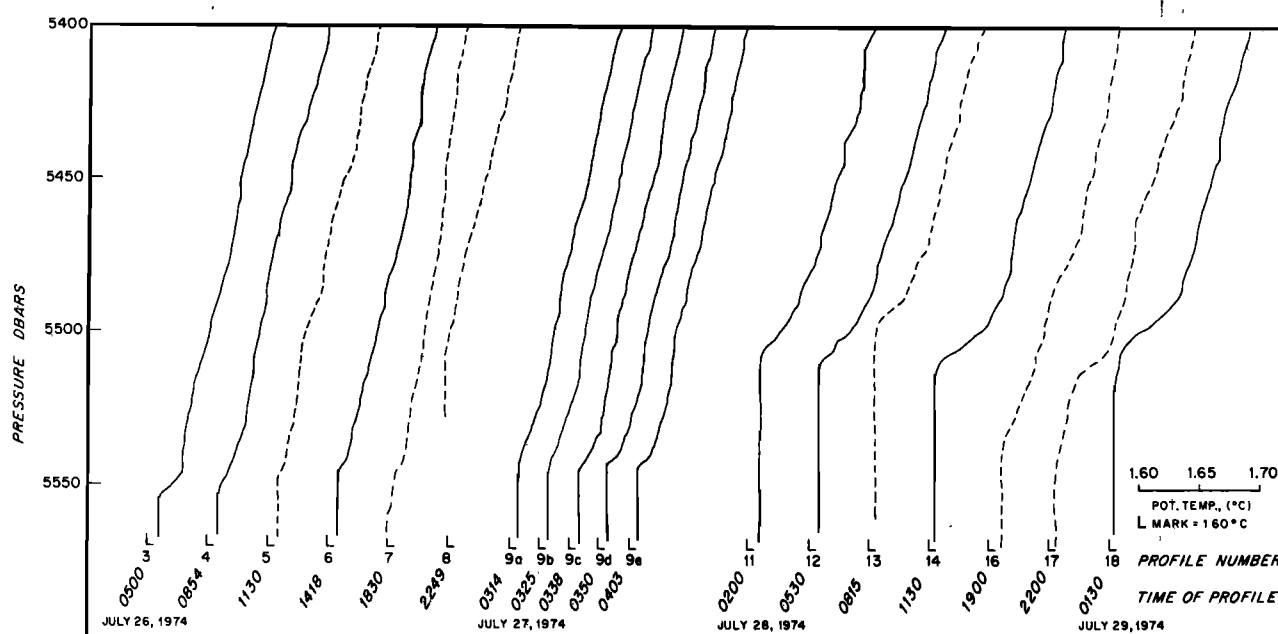


Fig. 4b. Profiles taken in region at edge of Hatteras Abyssal Plain, locations are shown in Figure 4a. Profiles taken within a 1-km radius of 28°03.6'N, 69°39'W are drawn with solid lines.

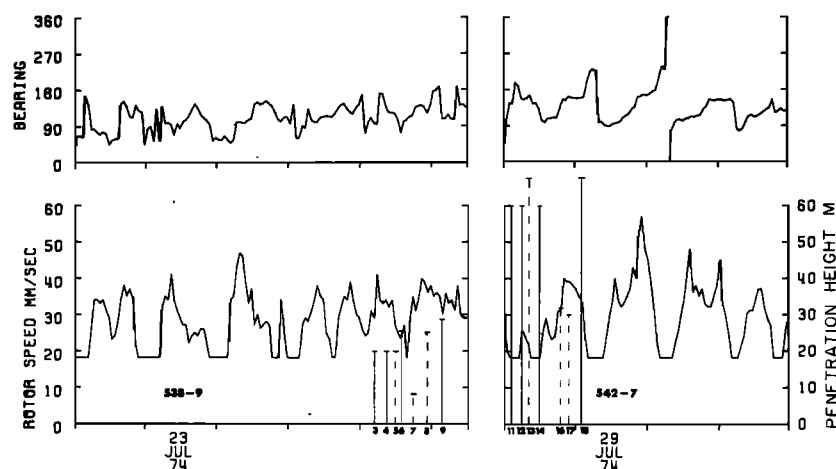


Fig. 4c. Penetration height, indicated by bars, versus time and current velocity and bearing versus time. Profiles taken within a 1-km radius are solid bars; locations are shown in Figure 4a.

constitute only the lower sixth of the well-mixed region of the bottom boundary layer. For the typical velocity of 4 cm/s the Ekman height is only 8 m.

The behavior of a turbulent boundary layer in high-frequency motions, here primarily inertial and tidal, is not understood. The effects of unsteadiness are discussed by *Wimbush and Munk* [1971], who note the time scale for the entire Ekman layer is $2\pi/f$; therefore those features of the boundary layer, in particular the logarithmic layer, with time scales very much less than $2\pi/f$ can be approximated by steady state theory. As suggested by *Munk et al.* [1970], the simplest procedure is to reinterpret the turbulent Ekman height (7) with u_* dependent on the local mean current. This was done here; however, even if u_* were dependent on the maximum value of the velocity, the Ekman layer height given by (8) would be larger by at most 6 m, since the most energetic inertial or tidal velocities are ~ 3 cm/s.

A model of the bottom boundary layer combining the effects of stratification and rotation has been proposed by *Thompson* [1973]. This is a slab model of the homogeneous well-mixed region, with the penetration height given by a bulk Richardson number closure assumption based on the height of the layer and jumps across the sharp interface of density and velocity. The velocity difference is that between the geostrophic velocity above and the mean velocity in the layer. This bulk Richardson number closure is also used by *Pollard et al.* [1973], to whom *Thompson* [1973] refers for a detailed explanation. The penetration height of 8 m for $U \sim 5$ cm/s, as suggested by *Thompson*, is much too small.

Csanady [1974] finds the parameterization of interfacial stability used by *Pollard et al.* [1973] and *Thompson* [1973] unattractive because the control of the mixed layer depth may be independent of the mechanism that maintains the stability of the interface. *Csanady* suggests that a limit to entrainment is set by the turbulent length and velocity scales in the mixed layer and the density differences across the interface. This asymptotic limit for the entrainment is deduced from the experiments of *Kato and Phillips* [1969]. The limit used by *Csanady* is given by

$$g'h/u_*^2 = 500 \quad (9)$$

where g' is the reduced gravitational acceleration at the interface, h is the layer thickness, and u_* is the friction velocity or

turbulent velocity scale. We note, however, that with $u_*^2 = 10^{-3}U^2$, used by *Csanady*, (9) is equivalent to a Froude number closure of $F = 1.4$, not very different from the value of unity chosen by *Pollard et al.* [1973] or the empirical result of $F \sim 1.7$ found here. The existence of an asymptotic entrainment limit, such as expressed by (9) above and used by *Csanady*, is certainly not established and must be questioned in light of the experiments of *Turner* [1973, p. 291], in which no tendency toward a limit is actually observed.

6. SUMMARY AND CONCLUSION

We have described various aspects of the bottom boundary layer as observed with a CTD profiler. Profiles can be divided into two distinct types. Those from the center of the Hatteras Abyssal Plain exhibit a structure that is characteristic of mixing a uniformly stratified region. Profiles from regions with rough or sloping topography show complicated structure. A profile from near the edge of the abyssal plain can exhibit either the simple or complicated structure. The results of a preliminary survey at the edge of the abyssal plain show spatial variability over 10 km and temporal variability over 3 days.

The characteristic well-mixed region of the bottom boundary layer does not form a pool or have a distinctive water mass characteristic. We have correlated the penetration height of this well-mixed region with the mean velocity to show that it is of dynamic origin. We believe the layer was formed by mixing of the stratified deep ocean.

The depth of the well-mixed region is large compared with semiempirical estimates of the turbulent Ekman layer height. The correlation of penetration height with velocity has allowed us to estimate the value of the internal Froude number of the layer as $F \sim 1.7$.

Acknowledgments. We are very grateful for the assistance of the WHOI buoy group with the collection and presentation of these results and for support by the Office of Naval Research under contract N000 14-66-C0241 NR 083-004 and by the International Decade of Ocean Exploration Office of the National Science Foundation (GX 29054). L. Armi is thankful for comments from Melbourne Briscoe, Gilles Corcos, Nelson Hogg, Ian Jones, and Jack Whitehead on an early version of the manuscript and for financial support as a postdoctoral scholar from the Woods Hole Oceanographic Institution.

This is contribution 3744 from the Woods Hole Oceanographic Institution, Woods Hole, Massachusetts.

REFERENCES

- Amos, A. A., A. L. Gordon, and E. D. Schneider, Water masses and circulation patterns in the region of the Blake-Bahama Outer Ridge, *Deep Sea Res.*, **18**, 145–165, 1971.
- Biscaye, P. E., and S. L. Eittreim, Variations in benthic boundary layer phenomena: Nepheloid layer in the North Atlantic Basin, in *Suspended Solids in Water*, edited by R. J. Gibbs, pp. 227–260, Plenum, New York, 1974.
- Brown, N. L., A precision CTD microprofiler, Proceedings of the IEEE International Conference on Engineering in the Ocean Environment, vol. 2, pp. 270–278, Inst. Elec. and Electron. Eng., New York, 1974.
- Bush, P. A., Bathymetry of the Mode-I region, *Mode Hot Line News*, **70**, 1, 1975.
- Bush, P. A., Bathymetry of the Mode region, *Deep Sea Res.*, in press, 1976.
- Caldwell, D. R., C. W. Van Atta, and K. N. Helland, A laboratory study of the turbulent Ekman layer, *Geophys. Fluid Dyn.*, **3**, 125–160, 1972.
- Cromwell, T., Pycnoclines created by mixing in an aquarium tank, *J. Mar. Res.*, **18**, 73–82, 1960.
- Csanady, G. T., On the 'resistance law' of a turbulent Ekman layer, *J. Atmos. Sci.*, **24**, 467–471, 1967.
- Csanady, G. T., Equilibrium theory of the planetary boundary layer with an inversion lid, *Boundary Layer Meteorol.*, **6**, 63–79, 1974.
- Eittreim, S., P. E. Biscaye, and A. F. Amos, Benthic nepheloid layers and the Ekman thermal pump, *J. Geophys. Res.*, **80**, 5061–5067, 1975.
- Fofonoff, N. P., S. P. Hayes, and R. C. Millard, Jr., WHOI/Brown CTD microprofiler: Methods of calibration and data handling, *Ref. 74-89*, Woods Hole Oceanogr. Inst., Woods Hole, Mass., 1974.
- Gibbs, R. J. (Ed.), *Suspended Solids in Water*, Plenum, New York, 1974.
- Gordon, A. L. (Ed.), *Studies in Physical Oceanography*, vol. 2, Gordon and Breach, New York, 1972.
- Gould, W. J., W. J. Schmitz, Jr., and C. Wunsch, Preliminary field results for a Mid-Ocean Dynamics Experiment (Mode-O), *Deep Sea Res.*, **21**, 911–931, 1974.
- Howroyd, G. C., and P. R. Slawson, The characteristics of a laboratory produced turbulent Ekman layer, *Boundary Layer Meteorol.*, **8**, 201–219, 1975.
- Kato, H., and O. M. Phillips, On the penetration of a turbulent layer into a stratified fluid, *J. Fluid Mech.*, **37**, 643–655, 1969.
- Knox, R. A., and R. C. Millard, Jr., Bottom sensing device for use with STD systems, *Deep Sea Res.*, **20**, 419–421, 1973.
- Linden, P. F., The deepening of a mixed layer in a stratified fluid, *J. Fluid Mech.*, **71**, 385–405, 1975.
- Millard, R. C., Jr., Bottom-layer observations from Mode and Iwex, *Mode Hot Line News*, **60**, 5, 1974.
- Mode I Dynamics Group, Dynamics and the analysis of Mode-I, The Mid-Ocean Dynamics Experiment—Mode-I, Mass. Inst. of Technol., Cambridge, 1975.
- Munk, W., F. Snodgrass, and M. Wimbush, Tides off-shore: Transition from California coastal to deep-seawaters, *Geophys. Fluid Dyn.*, **1**, 161–235, 1970.
- Pollard, R. T., P. B. Rhines, and R. Thompson, The deepening of the wind-mixed layer, *Geophys. Fluid Dyn.*, **3**, 381–404, 1973.
- Sanford, T. B., Observations of the vertical structure of internal waves, *J. Geophys. Res.*, **80**(27), 3861–3871, 1975.
- Thompson, R., Stratified Ekman boundary layer models, *Geophys. Fluid Dyn.*, **5**, 201–210, 1973.
- Turner, J. S., *Buoyancy Effects in Fluids*, Cambridge University Press, New York, 1973.
- Weatherly, G. L., A study of the bottom boundary layer of the Florida Current, *J. Phys. Oceanogr.*, **2**, 54–72, 1972.
- Weatherly, G. L., A numerical study of time-dependent turbulent Ekman layers over horizontal and sloping bottoms, *J. Phys. Oceanogr.*, **5**, 288–299, 1975.
- Weatherly, G. L., and P. P. Niiler, Bottom homogeneous layers in the Florida Current, *Geophys. Res. Lett.*, **1**, 316–319, 1974.
- Wimbush, M., and W. Munk, The benthic boundary layer, in *The Sea*, vol. 4, part 1, edited by A. E. Maxwell, pp. 731–758, John Wiley, New York, 1971.
- Worthington, L. V., and W. G. Metcalf, The relationship between potential temperature and salinity in deep Atlantic water, *Rapp. Proces Verb. Reunions Cons. Perma. Int. Explor. Mer*, **149**, 122–128, 1961.

(Received April 2, 1976;
revised June 21, 1976;
accepted June 21, 1976.)